

Unit 12, Lesson 1: Finding an Unknown Decimal in an Equation

Benchmark

MA.6.AR.2.4 Determine the unknown decimal or fraction in an equation involving any of the four operations, relating three numbers, with the unknown in any position.

Learning Target

- ☐ I can determine an unknown decimal in a numeric equation with three numbers.

Science, Technology, Engineering, and Mathematics (STEM) is a way of thinking and doing. STEM skills are useful to almost every career.

An individual does not need to be a master in all four areas of STEM to be successful; teamwork plays an important role. Playing to a person's strengths and working as a team creates a greater impact than working alone.

1. Describe a time where it was better for you to work as a team than alone.
2. List two jobs that involve working as a team but are in different fields of STEM.

12.1.2 Refresher: The Unknown Value

With your partner, complete the following.

In math, **variables** are used to represent unknown values.

In an algebraic expression, a **variable** is a letter that represents a number.

1. Circle the variable(s) in each expression.

$m + 5$

$2x + y$

$\frac{a}{4}$

Variables are found in algebraic equations.



An **equation** is a mathematical relation statement where two equivalent expressions or values are separated by an equal sign.

An **algebraic equation** is an equation with at least one variable.

2. Circle the algebraic equations.

$6 + 8 = 14$

$5y = 35$

$2w + 7 = 13$

$\frac{12}{3} + 8 = 12$



To **solve** an equation means to find the solution.

The **solution(s)** to an algebraic equation is (are) the value(s) of the variable(s) that makes the equation true.

12.1.3 Guided Instruction: Solving Equations Using Reasoning and Diagrams

For some equations, the solution can be found mentally or by using reasoning.

1. The solution to the equation $n - 12.1 = 35.3$ can be found using reasoning.
 - a. Discuss with your partner how to find the value of n in the equation $n - 12.1 = 35.3$.
 - b. The equation is rewritten vertically with a box in the place of the variable. Write the solution in the box and check whether it makes the equation true.

$$\begin{array}{r}
 \boxed{} \\
 - 12.1 \\
 \hline
 35.3
 \end{array}$$

Diagrams can also be helpful to find a solution.

2. Olivia gave equal amounts of money to each of her 3 children. Olivia gave each child \$16.75.

The bar model represents the equation $\frac{m}{3} = 16.75$, where m is the total amount of money Olivia gave all of her children.

16.75	16.75	16.75
m		

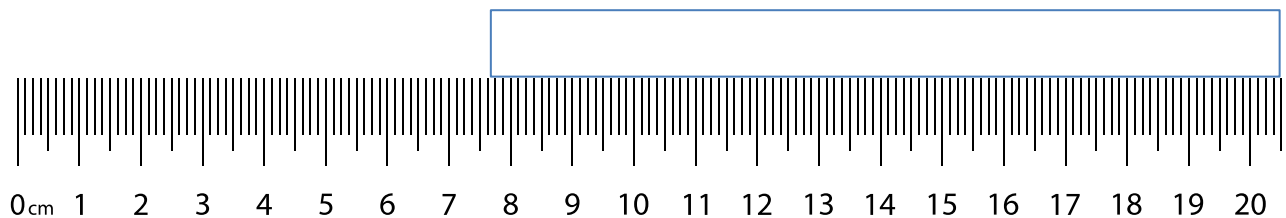
- a. How can the bar model be used to find the total amount of money Olivia gave all of her children, m ?

- b. What number can be divided by 3 to get a result of 16.75?

$$\frac{\boxed{}}{3} = 16.75$$

3. When five equal-sized paperclips are placed end-to-end, the total length is 16 centimeters. The equation $16 = 5p$ can be used to find the length of one paperclip, p .

The rectangle represents the total length of 5 paperclips.



- a. How can the picture be used to find the value of p in the equation $16 = 5p$?

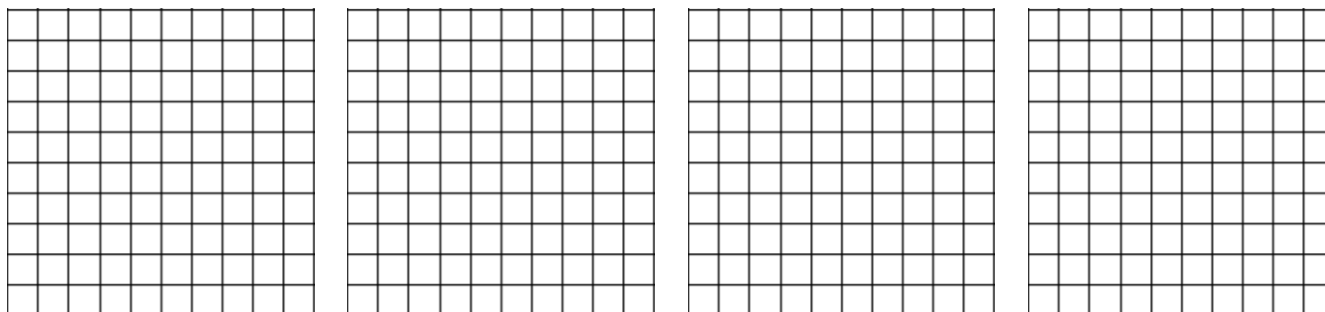
- b. What number can be multiplied by 5 to result in 16?

$$16 = 5(\boxed{})$$

- c. What is the length of each paperclip?

4. Myla challenged herself to walk 3.5 miles each day. This morning she walked 1.68 miles. This situation can be represented by the equation $3.5 = 1.68 + x$, where x is the number of miles Myla still needs to walk today to meet her goal.

Each 10 x 10 grid represents 1 mile.



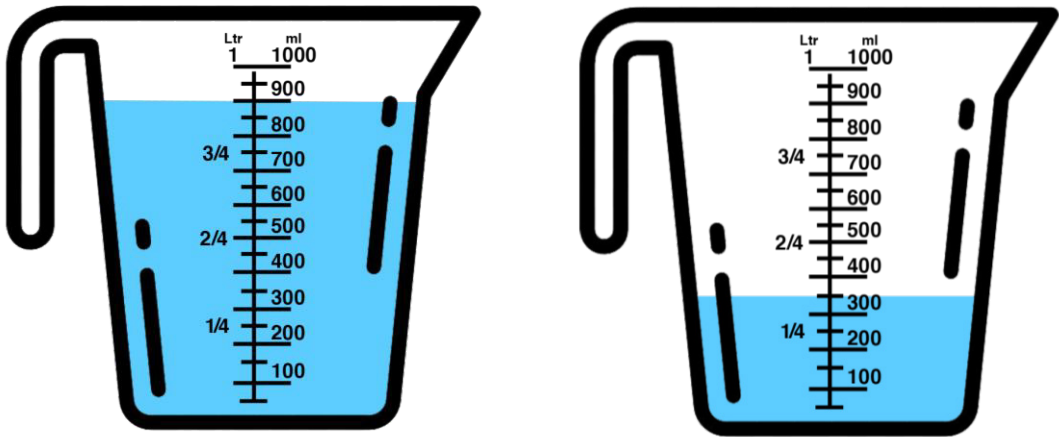
- a. Discuss with your partner how to use the 10 x 10 grids to represent the equation $3.5 = 1.68 + x$.
- b. What number can be added to 1.68 to get 3.5?
- c. Check your solution.

$$\begin{array}{r} 1.68 \\ + \\ \hline 3.50 \end{array}$$

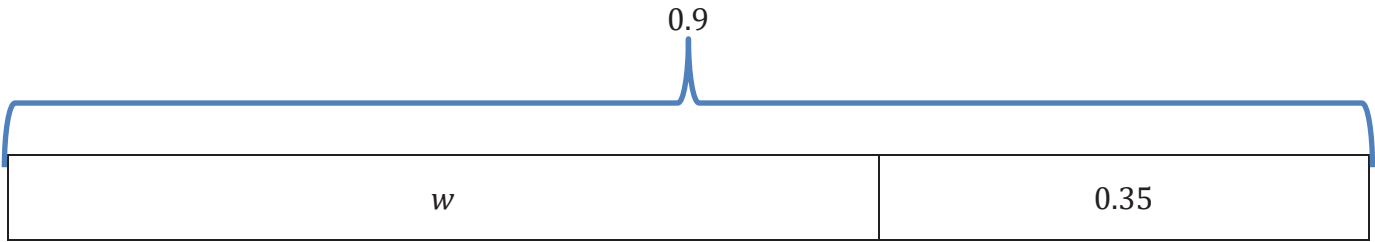
12.1.4 Collaboration: Solving Equations Using Reasoning & Diagrams

With your partner, solve the following problems using reasoning and diagrams.

1. Vincent had 0.9 liters (L) of water. He poured some into the mixing bowl and had 0.35 L left.



The bar model shown represents this situation.



- a. Discuss with your partner how the bar model represents Vincent’s situation. Then, write a brief summary of your discussion.
- b. The equation $0.9 - w = 0.35$ can also be used to represent this situation. Find the value of w , the amount of water Vincent poured into the mixing bowl.

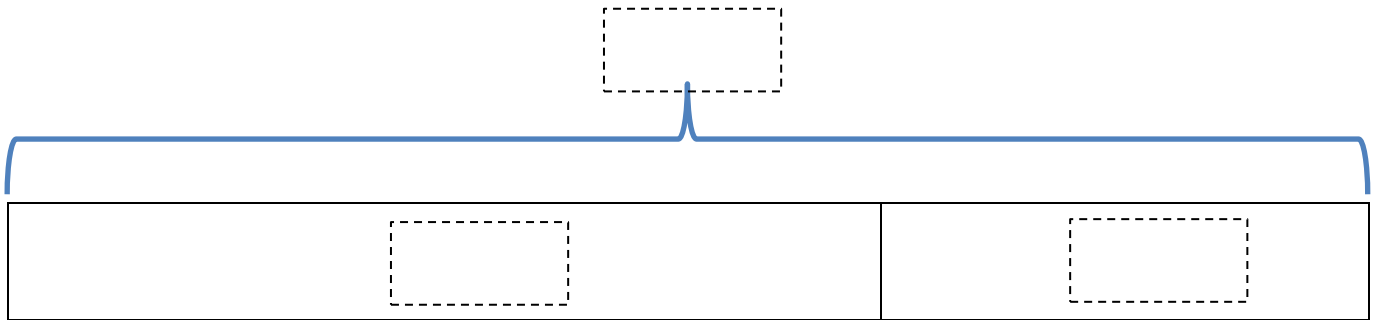
2. Josephine told her brother she would give him $\frac{1}{2}$ of the money they made from hosting a lemonade stand at their school carnival. She gave her brother \$24.25. This situation can be represented by the equation $\frac{1}{2}x = 24.25$ to find x , the amount of money they made at the lemonade stand.

- a. Use the bar model shown to represent the situation.

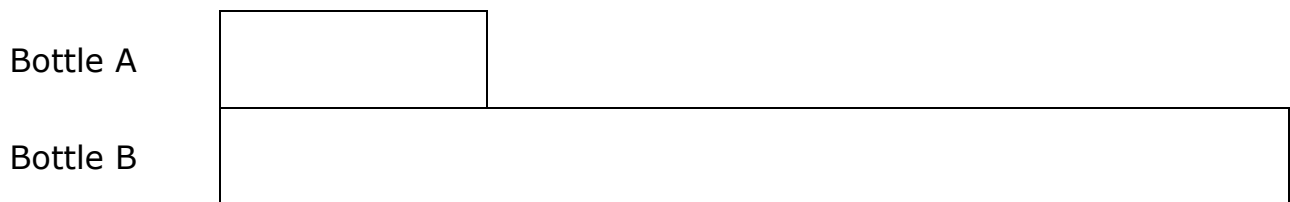
- b. Find the value of x , the amount of money they made at the lemonade stand.

12.1.5 Your Turn!

1. DeAndré paid \$58.54 for a game at Store A, which was \$9.21 less than the price of the game at Store B. This situation can be represented by the equation $58.54 = b - 9.21$, where b is the price of the game at Store B.
- a. Complete the bar model diagram to represent this situation.



- b. Find the value of b .
2. Bottle A contains 6 ounces of water, which is 25% of the amount of water in Bottle B. This situation can be represented by the equation $6 = 0.25w$, where w is the number of ounces of water in Bottle B.
- a. Complete the bar model to represent this situation.



- b. Find the value of w .

12.1.6 Wrap-Up: Reflection

1. Complete the learning pyramid.

One question I would like to have answered...	
One mistake I learned from today...	One thing I am not sure about...YET!
One thing I am confident I know is...	